

5 Chemical energetics

AS · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Exothermic reactions have negative enthalpy change; endothermic positive. Hess's law lets enthalpy changes be added because enthalpy is a state function. Bond energies approximate $\Delta H = \text{bonds broken} - \text{bonds formed}$; a calorimeter uses $q = mc\Delta T$. Quote signs and account for losses when evaluating experimental values.

Worked example: follow the reasoning

Burning fuel raises 100 g water by 20 K: $q \approx 100 \times 4.18 \times 20 = 8360 \text{ J}$ absorbed by water; reaction enthalpy is negative.

Quick check · multiple choice

A Which sign represents an exothermic ΔH ?

[1]

- A Negative
- B Positive always
- C Zero
- D Undefined

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 A 0.050 mol reaction warms 200 g water by 5.0 K. Estimate molar ΔH using $c = 4.18 \text{ J g}^{-1} \text{ K}^{-1}$. [3]

Final answer: _____

Answers and explanation

Quick check: A. Negative

Question 1 [3 marks]

$q = 200 \times 4.18 \times 5 = 4180 \text{ J [1]}$; $\Delta H = -4180/0.050 = -83\,600 \text{ J mol}^{-1} [1] = -83.6 \text{ kJ mol}^{-1} [1]$.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Chemistry 9701, 2025–2027. <https://www.cambridgeinternational.org/Images/664563-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.